

MERN STACK

DAY – 10

6 July 2026

JAVASCRIPT

Switch Statement

A switch statement is a selection control structure used to execute one block of code among many based on the value of an expression.

- It compares a variable with multiple case values.
- When a match is found, that block executes.
- break is used to exit the switch.
- default runs if no case matches.

Loops

A loop is a control structure used to execute a block of code repeatedly as long as a condition is true. Instead of writing the same code again and again, loops help automate repetition.

Types of loops

1. for loop: Used when the number of repetitions is known in advance.

- It has initialization, condition, and update in one line.
- Best for counting tasks.

```
console.log("SKIP MULTIPLE OF 7");
for (let i = 0; i <= 30; i++) {
  if (i % 7 == 0) {
    continue;
  }
  if (i == 25) {
    break;
  }
  console.log(i);
}
```

SKIP MULTIPLE OF 7

1
2
3
4
5
6
8
9
10
11
12
13
15
16
17
18
19
20
22
23
24

```

console.log("EVEN AND ODD NUMBERS");
for (let i = 0; i < 30; i++) {
  if (i % 2 == 0) {
    console.log("Even number", i);
  } else {
    console.log("Odd number", i);
  }
}

```

```

EVEN AND ODD NUMBERS
Even number 0
Odd number 1
Even number 2
Odd number 3
Even number 4
Odd number 5
Even number 6
Odd number 7
Even number 8
Odd number 9
Even number 10
Odd number 11
Even number 12
Odd number 13
Even number 14
Odd number 15
Even number 16
Odd number 17
Even number 18
Odd number 19
Even number 20
Odd number 21
Even number 22
Odd number 23
Even number 24
Odd number 25
Even number 26
Odd number 27
Even number 28
Odd number 29

```

2. while loop: Used when the number of repetitions is not fixed.

- First checks the condition.
- If true, executes the block.
- Repeats until condition becomes false.

```

console.log("WHILE LOPP");
let v=10;
while(v<20){
  console.log(v);
  v++;
}

```

```

WHILE LOPP
10
11
12
13
14
15
16
17
18
19

```

3. do-while loop: Similar to while loop, but:

- Executes the block at least once first
- Then checks the condition

```
console.log("DO-WHILE LOOP");
let q=1;
do{
    console.log(q);
    q++;
}while(q<10);
```

DO-WHILE LOOP

1
2
3
4
5
6
7
8
9

break Statement

break is used to immediately terminate execution of a loop or switch.

- In a loop → exits the entire loop.
- In a switch → exits the switch block.

Continue Statement

continue is used in loops to skip the current iteration and move to the next one.

- It does NOT stop the loop.
- It only skips remaining code for that iteration.

Functions

A function (called a method in Java) is a reusable block of code that performs a specific task.

- Helps avoid repetition.
- Improves modularity.
- Can take inputs (parameters) and return outputs.

```
<title>FUNCTIONS</title>
</head>
<body>
  <script>
    function name(){
      let a=10;
      let b=90;
      console.log(a+b);
    }
    name();

    function fun2(a,b){
      console.log(a+b)
    }
    fun2(10,20);
    fun2(80,90);
  </script>
</body>
```

30

Live reload enabled.

> `ctrl` `i` to turn on code suggestion

DOM (Document Object Model)

DOM is a programming interface for web documents (HTML/XML), used mainly in JavaScript.

- Represents the page as a tree structure.
- Each HTML element is a node in the tree.
- Allows programs to dynamically access and change content, structure, and styles of a webpage.

```

    div{
        margin-left: 650px;
    }
</style>
</head>

<body>
<h1 align="center">TASK 1</h1>
<div>

    <p> </p>
    <button onclick="html()">HTML</button>
    <button onclick="css()">CSS</button>
    <button onclick="java()">JAVASCRIPT</button>

</div>
<script>
    let a = document.getElementsByTagName("p")[0];
    function html() {
        a.innerHTML = "HTML";
    }
    function css() {
        a.innerHTML = "CSS";
    }
    function java() {
        a.innerHTML = "javascript";
    }

```

TASK 1

HTML

HTML CSS JAVASCRIPT

```

<style>
  div {
    margin-left: 700px;
  }
</style>
</head>

<body>
  <h1 align="center">TASK 2</h1>
  <div>

    <button onclick="inc()">INCREASE</button>
    <br><p align="left">0</p><br>
    <button onclick="dec()">DECREASE</button>
  </div>
  <script>
    let q = document.getElementsByTagName("p")[0];
    let i = 0;
    function inc() {
      i++;
      q.innerHTML = i;
    }
    function dec() {
      i--;
      q.innerHTML = i;
    }
  </script>
</body>

```

TASK 2

INCREASE

3

DECREASE

```

<body>
  <p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Aut, est consequuntur molestiae, dolorem soluta,
    laudantium dolor ea quo doloribus repudiandae nulla officiis autem eius voluptate ad explicabo ullam rem
    sapiente quae sequi iusto accusamus. Vero nesciunt modi sunt natus unde?</p>
  <button onclick="p()">Click me</button>
  <script>
    let q=document.getElementsByTagName("p")[0]
    function p(){
      q.innerHTML="Bye";
      q.style.backgroundColor="red";
    }
  </script>
</body>

```

Lorem ipsum dolor sit amet consectetur adipisicing elit. Aut, est consequuntur molestiae,
 dolorem soluta, laudantium dolor ea quo doloribus repudiandae nulla officiis autem eius
 voluptate ad explicabo ullam rem sapiente quae sequi iusto accusamus. Vero nesciunt modi
 sunt natus unde?

Click me

Bye

Click me